Original Article

# Seroprevalence of Cytomegalovirus Antibodies among Blood Donors in Aminu Kano Teaching Hospital, Kano, Nigeria

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## Abstract

**Introduction:** Transfusion transmissible infections (TTIs) may potentially lead to serious harm to recipients of blood transfusion. Cytomegalovirus (CMV) can be transmitted through blood transfusion, and its impact is the most severe among immunocompromised recipients. Its transmission can be reduced through screening of blood donors, blood and blood products for CMV antibodies, leuco-depletion or pathogen inactivation. **Objective:** This was a cross-sectional study conducted among 250 blood donors between October and December 2012 in Aminu Kano Teaching Hospital, Kano, Nigeria. To determine the prevalence of human immunodeficiency virus (HIV) I and II antibodies, HBsAg, hepatitis C virus (HCV) antibodies, syphilis and anti-CMV IgM and IgG antibodies. In addition, the type of donation and donation patterns were also assessed. **Materials and Methods:** Five milliliters were collected from the participants. One milliliter was dispensed into ethylenediaminetetraacetic acid (EDTA) container and mixed gently, the remaining 4 ml were dispensed into a plain bottle and was centrifuged; serum obtained was divided into two one was stored at − 20°C which was later used to test for the anti-CMV IgG/IgM antibodies, whereas the other portion was used for routine screening of TTIs. The EDTA sample was used to assess haemoglobin concentration. An interviewer administered simplified questionnaire was used to obtain information on demographic and other variables related to donation habits. Association between the socio-demographic characteristics of the blood donors and the anti-CMV antibodies was analysed. Similarly, association between the anti-CMV antibodies and the other routinely screened TTIs was assessed. Results were tested for evidence of statistical association. **Results:** Majority of the blood donors were males 242 (96.8%). One hundred and seven (42.6%) of the blood donors were within the age group of 25–34 years. Only 10.8% of the donors were voluntary donors whereas the remaining were family replacement donors. Prevalence rates of 100% and 4.4% were found for anti-CMV IgG and anti-CMV IgM antibodies, respectively. HBsAg, HIV I/II, anti-HCV antibodies and syphilis were prevalent in 6.8%, 1.2%, 0.8% and 0% of the donors, respectively. **Conclusion:** Seroprevalence of anti-CMV IgG and anti-CMV IgM antibodies among blood donors in this study was found to be 100% and 4.4%, respectively. Therefore, routine predonation screening for anti-CMV antibodies may not be cost effective. Leucocyte depletion and pathogen inactivation methods are better options. However, these can better be achieved when centralized system of blood transfusion is adopted. Therefore, there is a need to enhance the functional capacity of our national blood transfusion service.

**Keywords:** Blood donors, cytomegalovirus, IgM/IgG antibodies, Kano, leucocyte depletion, transfusion transmissible infections

## Introduction

Blood transfusion is an essential part of patient care. It saves lives and improves health. However, blood transfusion carries a potential risk of acute and delayed complications including transmission of infections.[1]

While therapeutic use of blood is a universal practice, the pattern of its utilisation greatly differs across the globe according to the country's level of health care development. Countries with sophisticated healthcare system utilise blood mainly for complex procedures like cardiac bypass surgeries, organ and stem cell transplantation as well as to care for trauma patients. Blood is also used to support patients undergoing treatment for various malignancies, especially haematologic malignancies. In the developing world, however, greater proportion of blood is used to treat women with obstetric complications and children suffering from severe anemia often resulting from malnutrition and malaria.[2]

Access this article online

Quick Response Code:

Website: www.njbcs.net

DOI: 10.4103/njbcs.njbcs_47_16

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**How to cite this article:** Gwarzo DH, Gwarzo AK, Ahmed SG. Seroprevalence of cytomegalovirus antibodies among blood donors in Aminu Kano Teaching Hospital, Kano, Nigeria. Niger J Basic Clin Sci 2017;14:8-14.

No matter the reasons for utilisation of blood, the safety of the recipient must be ensured. Of great concern are the transfusion transmissible infections (TTIs), which often present as late complications. TTIs were first associated with blood transfusions in the late 1940s.[3]

Cytomegalovirus (CMV) is one of the viruses that can be transmitted through blood transfusion. Human CMV is a host specific virus belonging to the herpes viridae family.[4] Similar to other herpes viruses, it has the ability to establish lifelong infections in its hosts and to undergo periodic reactivation. This property gives CMV the ability to remain dormant in blood donors with reactivation in the recipients of blood transfusion.[4,5] CMV poses serious health concerns as it may cause severe morbidity and/or mortality in congenitally infected newborns and immunocompromised patients, most notably transplant recipients and HIV infected persons.[4,5]

Although CMV has a worldwide distribution, it is more common in developing countries.[6] The immunocompromised population for whom CMV seronegative blood products are requested is increasing. This is because of the advances in medical care afforded to premature infants, establishment of more special baby care units (SCBU), and increase in facilities that support intrauterine transfusion. There is also a contribution from the use of immunosuppressive drugs for transplant recipients and cancer patients as well as the continuing spread of HIV infection and acquired immune deficiency syndrome (AIDS).[7]

With the successful take off of the renal transplant program in Aminu Kano Teaching Hospital (AKTH), there has been an increase in the demand for CMV-seronegative blood in our blood bank. Thus, there is a need to establish the prevalence of CMV among blood donors to make an informed decision on modifying the screening policy of donors to include routine CMV screening and leucodepletion.

Previous hospital-based studies done within and outside Nigeria have reported a lower prevalence of anti-CMV IgM antibodies (acute infection) compared to anti-CMV IgG antibodies (past exposure).[5,8] This study in addition to establishing the prevalence of IgM and IgG anti-CMV antibodies also assessed the presence of other TTIs such as HIV, Hepatitis B and C, and syphilis, which are routinely screened before blood donation. The study also assessed the types of blood donors and their donation habits.

## Materials and Methods

This study was conducted in the AKTH, Kano, North Western Nigeria. This study was a cross-sectional descriptive study and was conducted between October and December 2012.

### Study population and inclusion and exclusion criteria

All blood donors who presented to the blood donor clinic within the study period were consecutively recruited. Only those with normal blood pressure, pulse rate, and body temperature were enrolled. Similar to other parts of the world, commercial sex workers and individuals with a history of chronic illness as well as intravenous drug users are not accepted as blood donors. They were, therefore, excluded from the study.

### Sample size determination

The sample size was determined using the formula

$n = Z^2Pq/d^2$

where,

$n$ = Minimum sample size

z = 1.96 (confidence interval) which is a constant

P = Prevalence rate from a previous study (ratio of 19.5% in a study conducted in a tertiary center in Lagos among donors attending blood donor clinic in LUTH).[8]

Therefore, $P = 19.5\% = 0.195$

q = 1 − P = complimentary probability

$=1 - 0.195 = 0.805$

d = Precision (margin of error) at 95% confidence limit = 0.05

$n = (1.96)^2 \times 0.195 \times 0.805/(0.05)^2 = 241$

The minimum sample size was rounded up to 250, and hence, a sample size of 250 was used.

### Sampling method

Consecutive blood donors who consented to participate in the study, aged 18–60 years, weighed greater than 50 kg, with haemoglobin of ≥ 13.5 g/dl and 12.5g/dl for males and females, respectively, were recruited till the calculated sample size was achieved.

### Study instruments

Microwell plate, photometric reader of microplates, automatic microplates washing device for anti-CMV antibodies, hemocue Hb 301 haemoglobinometer and the interviewer administered questionnaire.

### Data collection and analysis

An interviewer administered questionnaire was used to obtain information on demographic characteristics and the donation habits. VDRL (for syphilis), hepatitis C antibodies, hepatitis B surface antigen, HIV antibodies, anti-CMV IgG and IgM antibodies were done. Haemoglobin was estimated using hemocue Hb 301 haemoglobinometer. The serum anti-CMV IgG and anti-CMV IgM antibodies were assessed using long incubation enzyme-linked immunosorbent assay (ELISA) techniques (DIA.PRO), whereas the VDRL test (CTK Biotech, Inc), hepatitis C antibodies (CLINOTECH-DIAGNOSTICS), hepatitis B surface antigen (CLINOTECH-DIAGNOSTICS) and HIV antibodies (DETERMINE) were done using rapid ELISA-based strips. Manufacturer instructions were strictly followed.

Five milliliters of whole blood were collected from the participants. One milliliter was dispensed into the EDTA container and mixed gently, and the remaining 4 ml were

dispensed into a plain bottle and was centrifuged, serum obtained was divided into two one was stored at − 20°C which was later used for CMV analysis, whereas the other portion was used for the routine screening of TTIs. The EDTA sample was used to assess haemoglobin concentration.

The data generated were collated, checked and analysed using a computer-based statistical package for the social services (SPSS) version 16.0. Frequencies and percentages were calculated for qualitative variables. For quantitative variables, means and standard deviations were determined. After this analysis, the association between variables was assessed. Simple logistic regression was also done to determine the odds ratio and 95% confidence interval. Level of significance was set at ≤0.05.

### Ethical consideration

Ethical clearance to conduct the study was obtained from the health research ethics committee of AKTH. Informed written consent was also obtained from the participants.

## Results

Two hundred and fifty blood donors were recruited to participate in the study. Table 1 shows the socio-demographic characteristics of the participants. Majority of the blood donors were males constituting 96.8% of the sample (242), only 3.2% (8) of the respondents were females with a male-to-female ratio of approximately 30:1. The mean age of the participants was 32.25 ± 8.8 years. Of the 250 blood donors, 242 (96.8%) were males, Islam was the predominant religion with Muslims constituting 239 (95.6%) of the donors; 225 (90%) were of the Hausa/Fulani ethnic group. Approximately half of the respondents had tertiary education constituting 124 (49.6%). There were 144 (57.6%) married participants and 106 (42.4%) single participants. Table 2 shows that family replacement donors were the majority 223 (89.2%), whereas voluntary non-remunerated donors constituted 27 (10.8%) of the blood donors.

The prevalence of anti-CMV IgM antibodies in this study was 4.4% (11), while 92.4% (231) tested negative, with 3.2% (8) having equivocal results. The prevalence of anti-CMV IgG antibodies in this study was 100% (250), none of the blood donors tested negative. This also shows that 11 (4.4%) of the blood donors are positive for both anti-CMV IgG and anti-CMV IgM antibodies signifying reinfection or reactivation of infection.

A total of 22 (8.8%) donors tested positive for either HIV, HBsAg or HCV. The prevalence of HIV I/II antibodies among the blood donors in this study was 1.2% (3), whereas 98.8% (247) were negative. The prevalence of HBsAg among the blood donors in this study was 6.8% (17), the remaining 93.2% (233) tested negative for HBsAg. The prevalence of anti-HCV antibodies in this study was 0.8% (2), and the remaining 99.2% (248) were negative. None of the participants was positive for syphilis [Table 3].

**Table 1: Socio-demographic characteristics of blood donors**

| Characteristic | *N*=250 | Percentage |
|---|---|---|
| Age group (years) | | |
| ≤ 24 | 48 | 19.1 |
| 24-35 | 107 | 42.6 |
| 35-44 | 68 | 27.1 |
| ≥45 | 27 | 11.2 |
| Marital status | | |
| Married | 144 | 57.6 |
| Single | 106 | 42.4 |
| Sex | | |
| Male | 242 | 96.8 |
| Female | 8 | 3.2 |
| Religion | | |
| Islam | 239 | 95.6 |
| Christianity | 11 | 4.4 |
| Ethnic group | | |
| Hausa/Fulani | 225 | 90 |
| Non Hausa/Fulani | 25 | 10 |
| Occupation | | |
| Civil servant | 64 | 25.6 |
| Business | 67 | 26.8 |
| Unskilled | 73 | 29.2 |
| Unemployed | 46 | 18.4 |
| Level of education | | |
| Non formal | 39 | 15.6 |
| Primary | 14 | 5.6 |
| Secondary | 73 | 29.2 |
| Tertiary | 124 | 49.6 |

**Table 2: Cultural and donation habits of blood donors**

| Characteristic | *N*=250 | Percentage |
|---|---|---|
| Previous blood donation | | |
| Yes | 119 | 47.6 |
| No | 131 | 52.4 |
| Type of blood donation | | |
| Voluntary | 27 | 10.8 |
| Family replacement | 223 | 89.2 |

Table 4 shows the association between anti-CMV IgM antibodies and socio-demographic characteristics of the blood donors. No significant statistical association was established between the anti-CMV IgM antibodies and the socio-demographic variables of the blood donors.

Table 5 shows the association between positivity of anti-CMV IgM antibodies and the donation habits of the blood donors. No significant statistical association was observed between the variables and the positivity of anti-CMV IgM antibodies. The proportion of family replacement donors who tested positive to anti-CMV IgM antibodies was 5.1%, whereas none of the voluntary donors tested positive for anti-CMV IgM antibodies 0%. However, no statistically significant association was found ($P$ = 0.617) between the two groups. The proportion of anti-CMV IgM antibodies positivity among the first time

donors was found to be 5.3% compared to the repeat donors, with a proportion of 3.9% ($P = 0.760$).

Table 6 shows the blood donors positive for both anti-CMV IgM antibodies and either HIV I/II antibodies, HBsAg or HCV antibodies. None of the HIV positive blood donors tested positive for IgM anti-CMV antibodies antibodies. Only 2 (11.8%) of the blood donors tested positive for both IgM anti-CMV antibodies and HBsAg. Of the 2 (0.8%) blood donors who tested positive for HCV antibodies, 1 tested positive for both HCV antibodies and IgM anti-CMV antibodies.

**Table 3: Prevalence of HBsAg, HIV I/II, antiHCV, anti-CMV and syphillis antibodies among blood donors**

| | *N*=250 | Percentage |
|---|---|---|
| Anti-CMV IgM | | |
| Positive | 11 | 4.4 |
| Negative | 231 | 93.4 |
| Equivocal | 8 | 3.2 |
| Anti-CMV IgG | | |
| Positive | 250 | 100 |
| Negative | 0 | 0 |
| HIV I/II | | |
| Positive | 3 | 1.2 |
| Negative | 247 | 98.8 |
| HBsAg | | |
| Positive | 17 | 6.8 |
| Negative | 233 | 93.2 |
| Anti-HCV | | |
| Positive | 2 | 0.8 |
| Negative | 248 | 99.2 |
| Syphilis | | |
| Positive | 0 | 0 |
| Negative | 250 | 100 |

## Discussion

TTIs may potentially lead to serious harm to the recipients of blood transfusion. The seroprevalence rate of anti-CMV IgG antibodies in this study is similar to earlier reports from Ibadan, south-western Nigeria and marginally higher than the rates reported from the south southern Edo state and southwestern Lagos state.[8,9] Ghana, a neighbouring West African country, reported a prevalence of 93%.[10] While studies from outside Africa documented prevalence rates of 93.3%, 95% and 96.4% in North Eastern Thailand, New Delhi and Brazil, respectively.[11-13] However, lower prevalence rate for the IgG anti-CMV antibodies were reported in the United States of America (USA) by Stavas *et al*. who reported a seroprevalence of 58.9% in a population of 6 years and above. Canon *et al*. (2010) had earlier associated racial and regional differences in the seroprevalence of CMV antibodies with higher rates seen in non-whites and in developing countries compared to whites and developed countries.[14,15]

Higher prevalence rate of IgM anti-CMV antibodies of 19.5% was reported in Lagos compared to the 4.4% reported in this study; however, Ojide *et al*. (2012) reported lower prevalence rate of 3.1% from Benin.[8,9] There is paucity of data on IgM anti-CMV antibodies among blood donors in Nigeria. Studies conducted in other developing countries documented rates of 12.2%, and 2.3% in Burkina Faso and Brazil, respectively.[13,16] While the prevalence rate reported in this study is within the range of previous studies from other parts of the country, there

**Table 4: Association between anti-CMV IgM antibodies and some socio-demographic characteristics of blood donors**

| Characteristic | Anti-CMV IgM Status | | |
|---|---|---|---|
| | Positive (*n* %) | Negative (*n* %) | OR (95% CI) |
| Age group (years) | | | |
| ≤ 24 | 1 (2.2) | 44 (97.8) | 1 |
| 24-35 | 4 (3.8) | 101 (96.2) | 1.174 (0.207-6.654) |
| 35-44 | 4 (6) | 63 (94) | 1.007 (0.161-6.283) |
| ≥ 45 | 2 (8) | 23 (92) | 0.511 (0.067-3.880) |
| Marital status | | | |
| Married | 8 (5.7) | 132 (94.3) | 1 |
| Single | 3 (3) | 99 (97) | 2 (0.517-7.732) |
| Sex | | | |
| Male | 10 (4.3) | 224 (95.7) | 1 |
| Female | 1 (12.5) | 7 (87.5) | 0.312 (0.035-2.789) |
| Ethnic group | | | |
| Hausa/Fulani | 10 (4.6) | 208 (95.4) | 1 |
| Non Hausa/Fulani | 1 (4.2) | 23 (95.8) | 1.105 (0.135-9.032) |
| Level of education | | | |
| Formal | 10 (4.9) | 193 (95.1) | 1 |
| Non forma | 1 (2.6) | 38 (97.4) | 1.968 (0.244-15.836) |

OR: Odds Ratio, CI: Confidence Interval. P for interaction age X IgM = 0.853. P for interaction marital status X IgM=0.294 . *P* for interaction sex X IgM=0.357. *P* for interaction ethnic group X IgM=0.924. *P* for interaction level of education X IgM=0.0.487

**Table 5: Association between IgM antibodies and donation habits**

| | Anti-CMV IgM Status | | *P* |
|---|---|---|---|
| | Positive (*n* %) | Negative (*n* %) | |
| Type of donation | | | |
| Voluntary | 0 (0) | 27 (100) | 0.760 |
| Family replacement | 11 (5.1) | 204 (94.1) | |
| Previous donation | | | |
| First time donors | 6 (5.3) | 108 (94.7) | 0.760 |
| Return donors | 5 (3.9) | 123 (96.1) | |

**Table 6: Prevalence of anti-CMV IgM antibodies among blood donors positive for HBsAg, HIV I/II and anti-HCV antibodies**

| | Anti-CMV IgM Status | |
|---|---|---|
| | Positive (*n* %) | Negative (*n* %) |
| HIV I/II antibodies | | |
| Positive | | 3 (100) |
| Negative | 0 (0) | 228 (95.4) |
| HBsAg | 11 (4.6) | |
| Positive | 2 (11.8) | 15 (88.2) |
| Negative | 9 (4) | 216 (96) |
| HCV antibodies | | |
| Positive | 1 (50) | 1 (50) |
| Negative | 10 (4.2) | 230 (95.8 |

is a need for a multicenter study to establish the true national prevalence.

In this study, there was a predominance of young blood donors in the age group of 25–34 years, which is a reflection of the demographic structure of developing countries. This study showed a predominance of male donors 442 (96.8%) with females constituting only 8 (3.2%). According to the tradition of a typical African society, there is a general misconception that women are not eligible to donate blood.[17] However, there was no statistically significant association between gender of the blood donors and IgM anti-CMV antibodies status ($P = 0.315$).

The study also showed that 57.6% of the blood donors were married, with those who were not married contributing to 42.4% of the blood donors. However, we did not find any statistically significant association between marital status of the blood donors and anti-CMV IgM antibodies status ($P = 0.365$).

The finding of only 10.8% voluntary non-remunerated donors (VNRDs) calls for concern. Although this has shown an improvement over the previous report of 3.6%,[18] it shows a wide variation from the recommended 100% VNRDs by the WHO. This improvement may have been achieved by the provision of a stand-alone donor center within the hospital, a dedicated blood donor organizer and ambulance, as well as increase in public enlightenment and blood donation campaigns. Though family replacements donors have continued to contribute greater than 80% of blood donation, there is still the need for increased government commitment and funding in order to meet the WHO standards. Expectedly none of the donors was a commercial or a paid donor because of the strict measures taken by the hospital management against commercial donation. However, it is also possible that commercial or paid donors may have presented in the guise of family replacement donors as earlier reported in India following legislation that bans commercial blood donation.[19] These figures are better than the ones reported by the Federal ministry of health in which commercial donors constituted 25% of blood donors whereas 75% were family replacement donors in Nigerian public hospitals.[1,20] The study further revealed that irrespective of the type of donation, 47.6% of the blood donors were repeat donors, whereas 52.4% were first time donors. This pattern indicates good donor mobilization and retention, which may also partly account for the lack of commercial/paid donors reported in this study. Although none of the VNRDs tested positive for anti-CMV IgM antibodies as compared to 11% of the family replacement donors who tested positive for the antibodies, we did not find any statistically significant association between type of donation and anti-CMV IgM status ($P = 0.617$). However, this is in tandem with previous reports that have shown that voluntary donors are safer than family replacement donors in terms of transmission of infection.[1,3,21] The frequency of donation did not appear to have statistically significant association with anti-CMV IgM status among first time or repeat donors probably because CMV antibodies are not routinely screened for in our blood donor clinic. Repeat donors are usually safer than first time donors with regards to TTIs because they have had previous counseling and testing, are informed of risk factors, and are likely to keep safe. Repeat donation may also help in eliminating TTIs that are not detected at the first donation because of the window period, especially where plasma is quarantined and used only after the donor has represented and tests negative.[1,2]

The prevalence of HIVI/II antibodies and HCV antibodies reported in this study (1.2% and 0.8%, respectively) is comparable to that reported earlier by Nwanko *et al*. who reported prevalence rates of 1.4% and 1.8%, respectively, from another hospital in the same state.[22] However, the rates for HBsAg and syphilis were slightly higher in the earlier study (11.1% and 7.5%, respectively) compared to those reported in this study; 6.8% and 0% for HBsAg and syphilis, respectively. This may be related to the type of donors who usually patronize their study center, which runs free services and consequently is likely to attract patients from lower socio-economic class. Therefore, it is expected that significant proportion of the blood donors from general hospital (the site of the previous study) are likely to be first time donors with no previous pre-donation counseling and testing for TTIs. Similar comparable prevalence rates for HIV antibodies and syphilis were reported to be 0.7% and 1.2%, respectively, among blood donors in a tertiary center in North Eastern Nigeria.[23,24] Salawu

*et al*. (2010) reported similar prevalence rates for TTIs among blood donors in another tertiary health care facility in Ile-Ife, Nigeria over the period of 3.5 years; prevalence rates of 7.5%, 0.96%, 0.86% and 2.6% for HBsAg, anti-HIV, anti-HCV and VDRL, respectively were reported.[25] Studies conducted outside the country have also reported similar trends in the prevalence of TTIs among blood donors. A study done in India, a developing country like Nigeria, reported prevalence rates of 0.6%, 1.7%, 0.8% and 0.7% for HIV, HBsAg, HCV and syphilis, respectively, similar to the ones we reported in this study.[26] We did not establish any significant statistical association between the prevalence of the routinely screened TTIs and anti-CMV IgM status of the blood donors. None of the blood donors was positive for both HIV antibodies and anti-CMV IgM antibodies. Among the blood donors with positive HBsAg, 11.8% were also positive for anti-CMV IgM antibodies. Fifty percent of the blood donors who were positive for HCV antibodies were also found to be positive for anti-CMV IgM antibodies, however, no statistically significant association was documented ($P = 0.89\%$).

### Limitation of the study

1. Majority of the donors were males due to the already established pattern of blood donation in our blood bank. Results cannot, therefore, be generalised.
2. Blood donors with equivocal results of IgM anti-CMV antibodies could not be recalled for re-testing with fresh samples because of logistic reasons.

## Conclusion

Seroprevalence of anti-CMV IgG and anti-CMV IgM antibodies among blood donors in this study was found to be 100% and 4.4%, respectively. We recommend screening for anti-CMV IgM antibodies for immunocompromised recipients such as neonates, post-transplant patients, chemotherapy and other patients on regular blood transfusion. Other alternatives of protecting the recipient from CMV infection such as pathogen in activation and leucocyte depletion will be the better options for anti-CMV IgG positive blood. However, this cannot be fully achieved without a fully functional centralized transfusion service.

Among the routinely screened for TTIs, prevalence rate of HBsAg was the highest with 6.8%, and fortunately, this can be prevented through immunization. Therefore, mass immunization of the populace, especially health workers, should be encouraged. Efforts should be made to convert these repeat donors to VNRDs who have been shown to be the safest in terms of transmission of infections.

### Recommendation

1. There is a need to augment the functional capacity of the national blood transfusion service to be able to, among other things, provide leucocyte depleted or pathogen inactivated blood and/or products to the increasing population of immunocompromise Nigerians.
2. Blood donor recruitment campaigns should be done with the aim of recruiting and retaining VNRDs.
3. Studies need to assess the efficacy of prophylactic CMV treatment for patient planned for organ or stem cell transplant.

### Financial support and sponsorship

Nil.

### Conflicts of interest

There are no conflicts of interest.

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